



**THE UNIVERSITY OF QUEENSLAND
SCHOOL OF CIVIL ENGINEERING
CIVL4333 Advanced Concrete Design
Assignment 3**

A simply-supported prestressed concrete beam with a total span of 20 m is designed to carry a permanent uniformly distributed superimposed dead load of 2 kN/m and live load of 20 kN/m in addition to its self-weight. The beam is post-tensioned by single parabolic cable, which consists of 14 strands of nominal diameter 15.2 mm. The maximum eccentricity at mid-span of the cable is indicated in Fig 1. The cable profile is symmetric about the mid-span as shown in Fig 1. The cable is stressed initially to 75% of the characteristic breaking load. The material and section properties are as follows:

Assume immediate losses in prestress force to be 12%, and deferred losses to be 6%.

Concrete

- Characteristic concrete strength = 32 MPa (28th day); 25 MPa (at transfer)
- Young's Modulus = 30,100 MPa (28th day); 26100 MPa (at transfer)
- Density of concrete (with steel) = 25 kN/m³
- take $\varepsilon_{cs} = 545 \times 10^{-6}$ (tb 3.1.7.2 of AS3600, $t=30$ years, temperate inland)

Prestressing strands nominal diameter 15.2 mm

- Characteristic breaking load = 250 kN
- Nominal area of one strand = 143 mm²
- Young's Modulus of strand $E_p = 190,000$ MPa
- Characteristic yield strength, $f_{py} = 1790$ MPa
- Duct diameter = 112mm
- $d_p = 550 + 450 = 1000$ mm

Cross section properties

- Area of concrete $A_g = 363,000$ mm²
- Second moment of area (moment of inertia) $I = 62 \times 10^9$ mm⁴
- Centroid = 650 mm measured from bottom
- Hypothetical thickness $t_h = 177$ mm
- Area of non-prestressing steel $A_s = 1,250$ mm², $f_{sy} = 500$ MPa
- Shear reinforcement R10 $f_{sy,f} = 250$ MPa, area of 1xR10 = 80mm²
- $d_{st} = 1150$ mm

Take strength reduction factor ϕ for flexure = 0.85 and ϕ for shear = 0.75

- Determine if the beam is cracked under factored service loads?
- Check the ultimate bending moment capacity (ϕM_u) at mid-span.
- Check the ultimate shear capacity (ϕV_u) of the beam at 1.0 m from either support and determine shear reinforcement requirements.

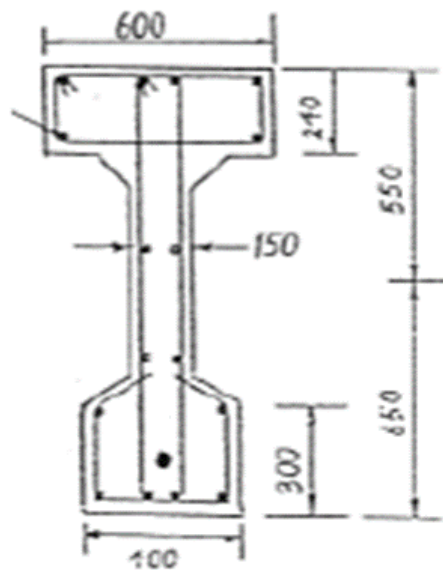
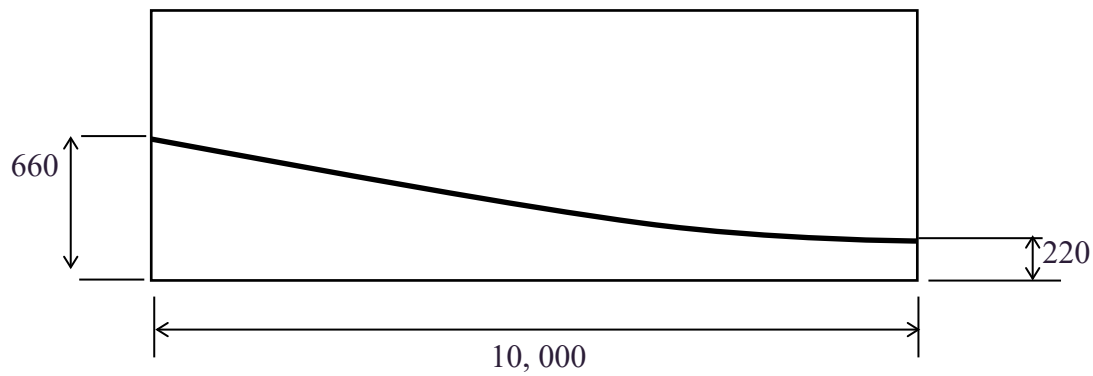


Fig 1 Elevation of beam showing prestressing steel profile (half span only) and cross section at midspan

END OF ASSIGNMENT 3